Visualize the data using Python libraries matplotlib, seaborn by plotting the graphs for assignment no. Air quality.

In [1]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

In [5]:
```
df = pd.read_csv('data_airquality.csv', encoding='latin1')
```

```
C:\Users\Rohit\AppData\Local\Temp\ipykernel_20556\3393981300.py:1: DtypeWarning:
Columns (0) have mixed types. Specify dtype option on import or set low_memory=Fa
lse.
  df = pd.read_csv('data_airquality.csv', encoding='latin1')
```

In [6]:
```
df = pd.read_csv('data_airquality.csv', encoding='latin1', low_memory=False)
```

In [7]:
```
df.head()
```

Out[7]:

	stn_code	sampling_date	state	location	agency	type	so2	no2	rspm
0	150	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	4.8	17.4	NaN
1	151	February - M021990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	3.1	7.0	NaN
2	152	February - M021990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.2	28.5	NaN
3	150	March - M031990	Andhra Pradesh	Hyderabad	NaN	Residential, Rural and other Areas	6.3	14.7	NaN
4	151	March - M031990	Andhra Pradesh	Hyderabad	NaN	Industrial Area	4.7	7.5	NaN

In [8]:
```
print(df.columns)
```

```
Index(['stn_code', 'sampling_date', 'state', 'location', 'agency', 'type',
       'so2', 'no2', 'rspm', 'spm', 'location_monitoring_station', 'pm2_5',
       'date'],
      dtype='object')
```

In [9]:
```
df.dtypes
```

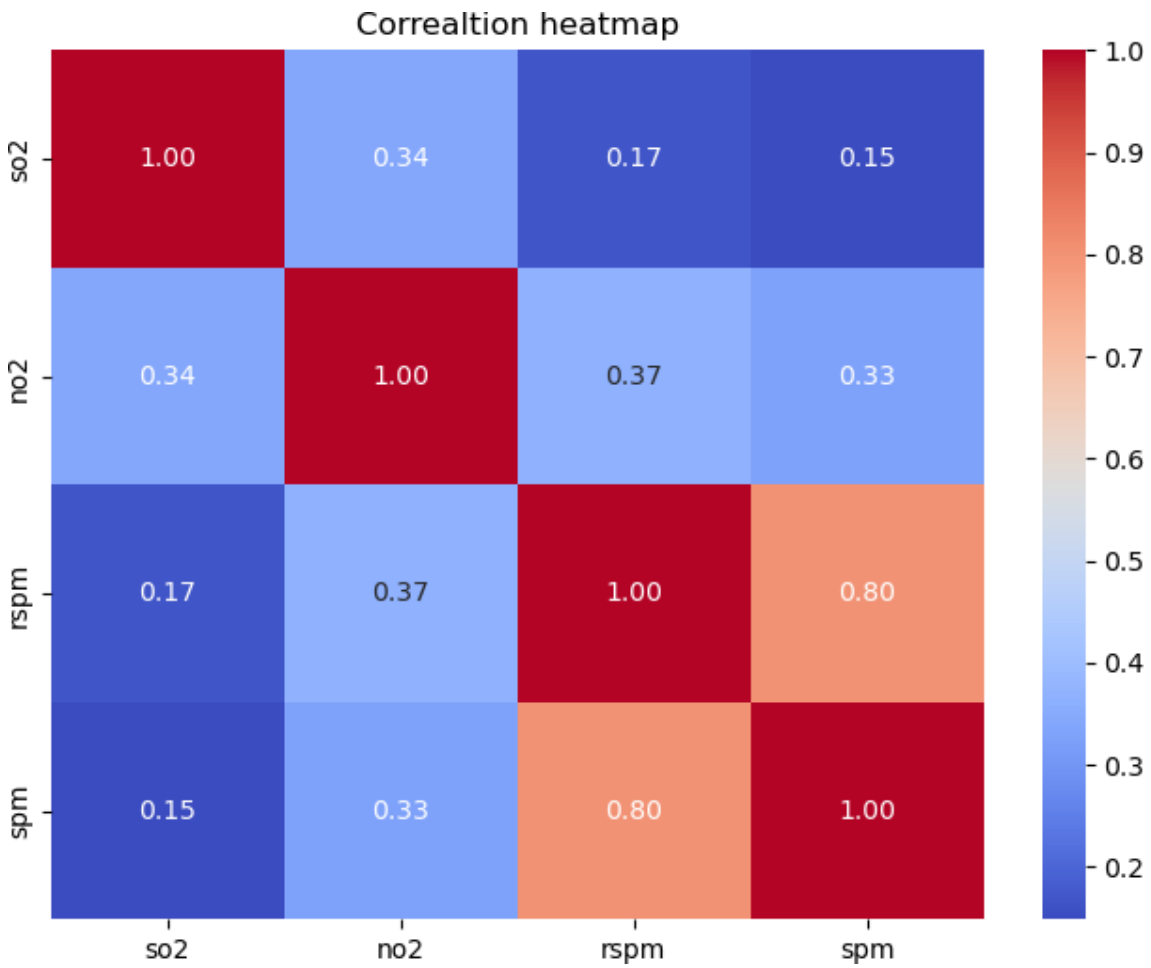
Out[9]:
```
stn_code                        object
sampling_date                   object
state                           object
location                        object
agency                          object
type                            object
so2                             float64
no2                             float64
rspm                            float64
spm                             float64
location_monitoring_station     object
pm2_5                           float64
date                            object
dtype: object
```

In [10]:
```
df = df.apply(pd.to_numeric,  errors='coerce')
```

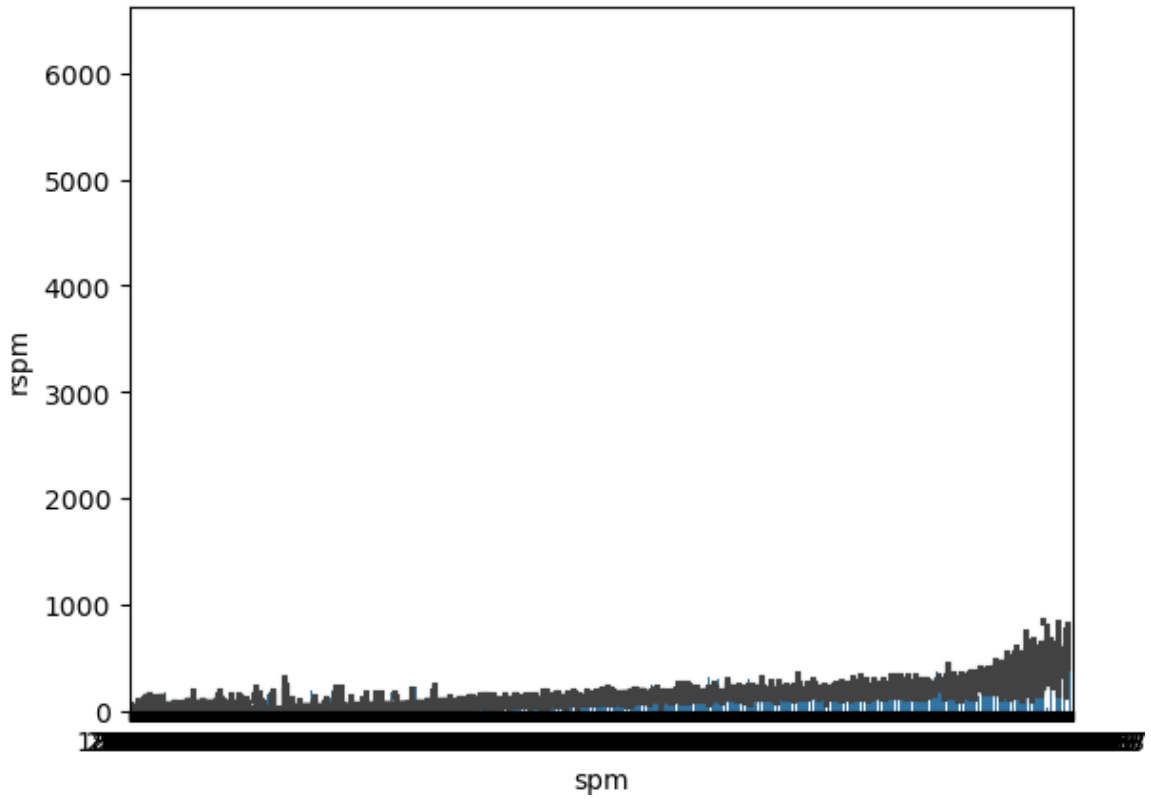
In [11]:
```
df.dtypes
```

Out[11]:
```
stn_code                        float64
sampling_date                   float64
state                           float64
location                        float64
agency                          float64
type                            float64
so2                             float64
no2                             float64
rspm                            float64
spm                             float64
location_monitoring_station     float64
pm2_5                           float64
date                            float64
dtype: object
```

In [26]:
```
plt.figure(figsize=(8,6))
cols = ['so2', 'no2', 'rspm',  'spm']
corr = df[cols].corr()
sns.heatmap(corr,  annot=True,  cmap="coolwarm",  fmt=".2f")
plt.title("Correaltion heatmap")
plt.show()
```

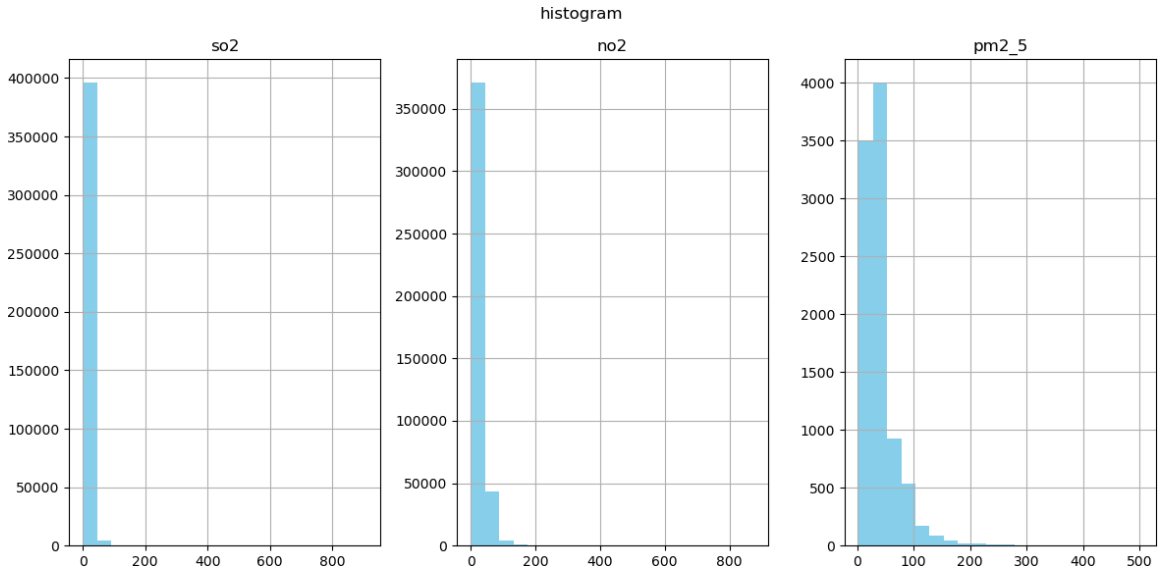


In [27]:
```
sns.barplot(x='spm', y='rspm', data=df)
plt.show()
```

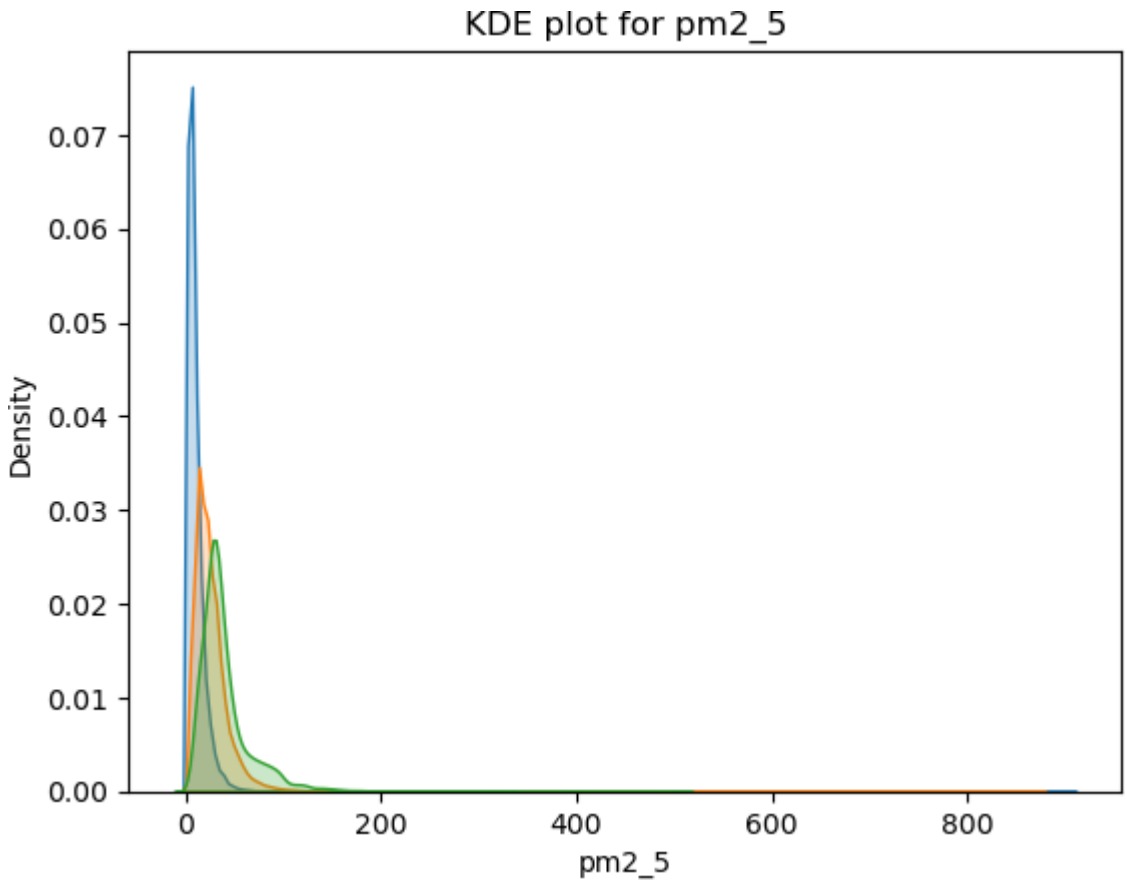


In [29]:
```
#Histogram
df[["so2","no2","pm2_5"]].hist(bins=20, figsize=(12,6), layout=(1,3), color='sky
```

```
plt.suptitle("histogram")
plt.tight_layout()
plt.show()
```

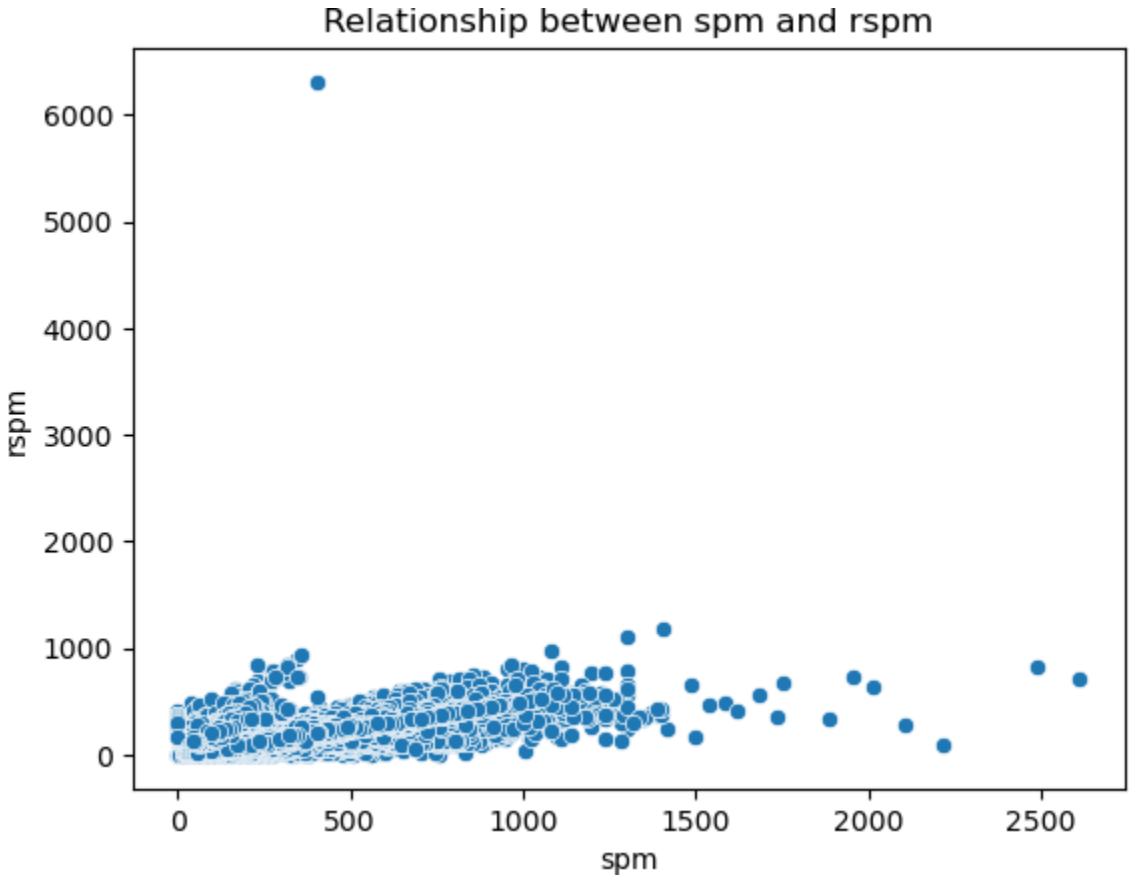


In [31]:
```
for column in ["so2","no2","pm2_5"]:
    sns.kdeplot(data=df[column], fill=True)
    plt.title(f"KDE plot for {column}")
    plt.xlabel(column)
    plt.ylabel("Density")
plt.show()
```

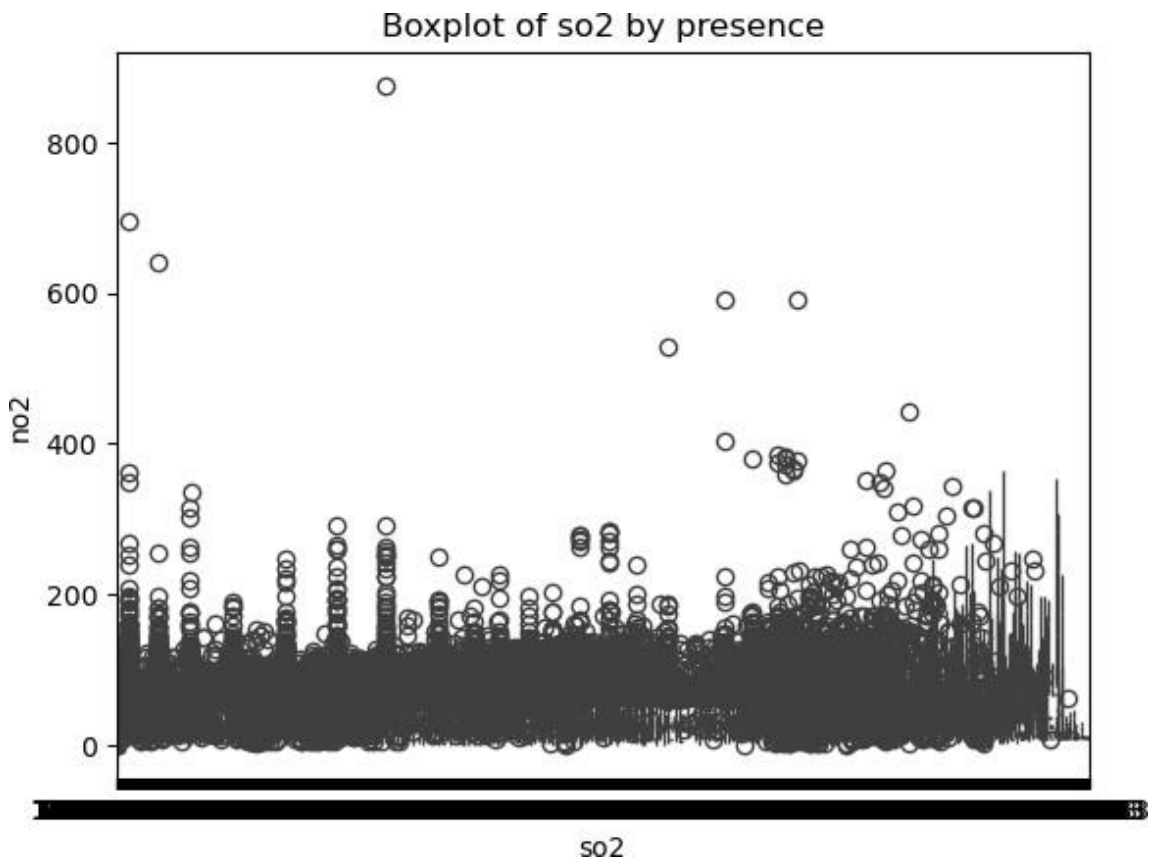


In [32]:
```
sns.scatterplot(x='spm', y='rspm', data=df)
plt.xlabel("spm")
plt.ylabel("rspm")
```

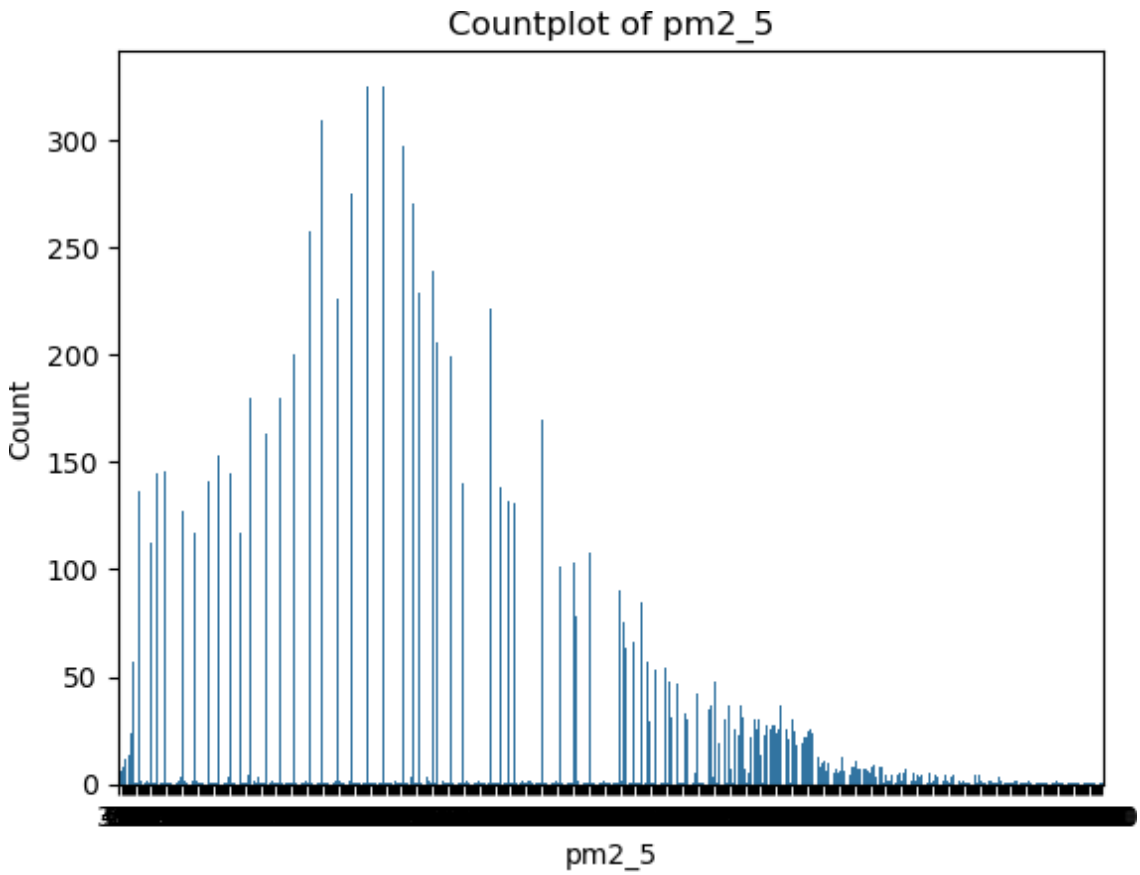
```
plt.title("Relationship between spm and rspm")
plt.show()
```



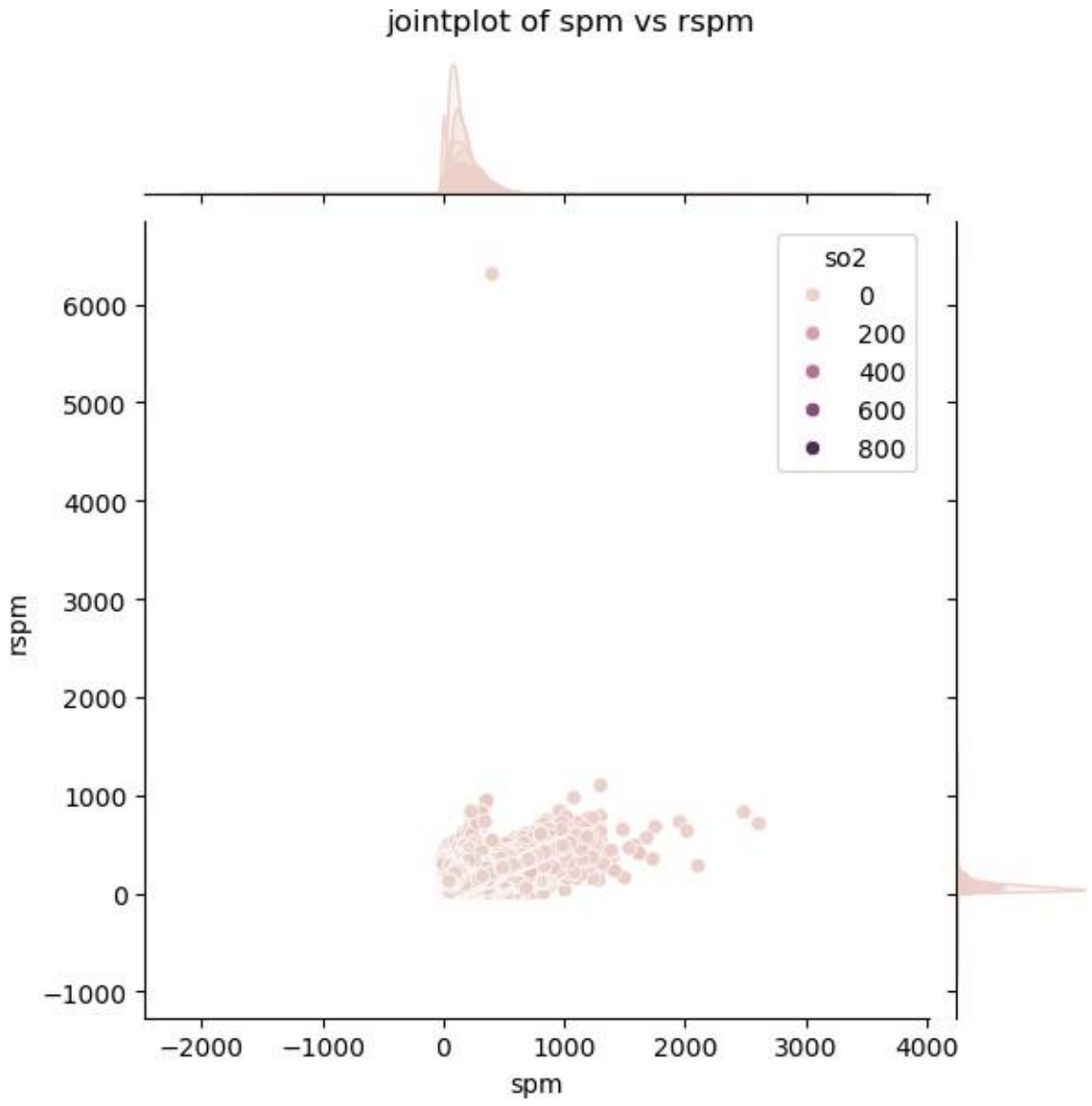
In [33]:
```
sns.boxplot(x='so2', y='no2', data=df)
plt.title("Boxplot of so2 by presence")
plt.show()
```



In [34]:
```
sns.countplot(x='pm2_5',  data=df)
plt.title("Countplot of pm2_5")
plt.xlabel("pm2_5")
plt.ylabel("Count")
plt.show()
```



In [35]:
```
sns.jointplot(data=df, x='spm', y='rspm', kind='scatter', hue='so2')
plt.suptitle('jointplot of spm vs rspm',y=1.02)
plt.show()
```



In [ ]: